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United States Patent	12405945
Kind Code	B2
Date of Patent	September 02, 2025
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Searches of highly structured data

Abstract

Techniques related to searches of highly structured data are described. A body of data may be represented by an object-centric data model. For a search of the body of data, an indication of a particular search template to use may be received. The particular search template may specify one or more hierarchical object types that are within a scope of the search. The one or more hierarchical object types may be defined in the object-centric data model. The particular search template may specify at least one search field. A user interface may be generated based on the particular search template. The user interface may include the at least one search field.

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Appl. No.: 16/408885

Filed: May 10, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20190266153 A1	Aug. 29, 2019

Related U.S. Application Data

continuation parent-doc US 14875536 20151005 US 10296617 child-doc US 16408885

Publication Classification

Int. Cl.: **G06F7/00** (20060101); **G06F3/0484** (20220101); **G06F16/242** (20190101); **G06F16/25** (20190101); **G06F16/28** (20190101)

U.S. Cl.:

CPC **G06F16/2428** (20190101); **G06F3/0484** (20130101); **G06F16/2423** (20190101); **G06F16/252** (20190101); **G06F16/289** (20190101);

Field of Classification Search

CPC: G06F (16/2423); G06F (16/252); G06F (16/2428); G06F (3/0484); G06F (16/289)

USPC: 707/766; 707/779; 707/999.005; 707/999.003; 707/706; 707/722; 707/769; 707/802; 707/E17.005; 707/E17.006

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS; BENEFIT CLAIM (1) This application claims the benefit under 35 U.S.C. § 120 as a Continuation of application Ser. No. 14/875,536, filed Oct. 5, 2015, which incorporates by reference application Ser. No. 13/831,199, filed Mar. 14, 2013 (now U.S. Pat. No. 10,037,314), the entire contents of which are hereby incorporated by reference for all purposes as if fully set forth herein. Applicants hereby rescind any disclaimer of claim scope in the parent application(s) or the prosecution history thereof and advise the USPTO that the claims in this application may be broader than any claim in the parent applications.

TECHNICAL FIELD

(1) The present Application relates to information retrieval technology. More specifically, the example embodiment(s) described below relate to searches of highly structured data.

BACKGROUND

(2) The approaches described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section.

(3) Computers are very powerful tools for searching information. A search engine is a common mechanism that allows users to search for information using computers. A search engine accepts a search query as input. A search query is typically composed of one or more keywords and provides a search result as output. The search result identifies information that the search engine has determined “satisfies” the search query. Search engines often maintain an index of a corpus of information that allows the search engine to efficiently identify information in the corpus that satisfies a given search query.

(4) One type of well-known search engine is an Internet search engine. Internet search engines are useful for searching semi-structured or unstructured data, such as the text content of a web page. However, the user interfaces of Internet search engines typically take a “one size fits all” approach with regard to how the user may scope the search. In particular, Internet search engines typically provide only a single text entry field into which the user enters one or more keywords. The Internet search engine then uses the entered keywords to identify information items that satisfy the entered keywords. This approach works well with the semi-structured and unstructured data that is indexed by Internet search engines, because such data typically is not represented by a highly-structured data model that is known to the user a priori the search. However, for highly structured data represented by a data model that the user has knowledge of prior to the search, the limited user interfaces provided by Internet search engines may be inadequate, inefficient, or cumbersome for users.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The example embodiment(s) of the present Application are illustrated, by way of example and not limitation, in the accompanying drawings in which like reference numerals refer to similar elements and in which:

(2) FIG. 1 depicts an example object-centric data model.

(3) FIG. 2 depicts an example ontology.

(4) FIG. 3 depicts an example user interface.

(5) FIG. 4 depicts a plurality of example search templates.

(6) FIG. 5A, FIG. 5B, FIG. 5C depict example approaches for obtaining input to search fields.

(7) FIG. 6 depicts an example customizable format for search results.

(8) FIG. 7, FIG. 8, FIG. 9 depict example detailed views of a search result.

(9) FIG. 10 is a flow diagram that depicts an approach for searching highly structured data, according to some example embodiments of the present invention.

(10) FIG. 11 depicts an example computer system in which embodiments may be implemented.

(11) FIG. 12 is a very general block diagram of a computing device in which the example embodiment(s) of the present Application may be embodied.

(12) FIG. 13 is a block diagram of a basic software system for controlling the operation of the computing device.

DESCRIPTION OF THE EXAMPLE EMBODIMENT(S)

(13) In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the example embodiment(s) of the present Application. It will be apparent, however, that the example embodiment(s) may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the example embodiment(s). Modifiers such as “first” and “second” may be used to differentiate elements, but the modifiers do not necessarily indicate any particular order.

(14) General Overview

(15) Computer-implemented techniques for helping a user search a body of data that is highly structured are described. In some example embodiments, the body of data is represented by an object-centric data model. The object-centric data model is centered on the notion of data objects and properties of the data objects. Furthermore, the object-centric data model is based on an ontology that defines hierarchical object types and property types. For example, a data object in the body of data may have the hierarchical object type “Employee” and have a “Name” property, a “Title” property,” and a “Salary” property. The ontology may define the “Employee” object type as a child object type of the “Person” object type. Thus, the “Employee” object is also a “Person” object according to the object-centric data model.

(16) In some example embodiments, the techniques encompass a computer-implemented method performed at one or more computing devices. The one or more computing devices include one or more processors and storage media storing one or more computer programs executed by the one or more processors to perform the method.

(17) Performance of the method includes performing the operation of receiving an indication of a search template to use for a search of the body of data represented by the object-centric data model. The search may have a scope specified by the template. In particular, the search template may specify one or more hierarchical object types, defined in the object-centric data model, that are within the scope of the search. The search template may also specify a plurality of search fields.

(18) Performance of the method may further include performing the operations of: based on the search template, generating a user interface that includes the plurality of search fields; and providing the user interface to a user. For at least a couple of reasons, the user interface may allow a user to search the body of data more efficiently and with higher precision and recall. One reason is that searches of the body of data initiated via the user interface may be scoped according to the one or more hierarchical object types specified in the search template. Scoping a search may involve associating a set of search fields with the one or more hierarchical object types that are specified. Another reason is that the search fields presented to the user in the user interface may be limited to those that are relevant to the one or more hierarchical object types specified in the search template. In other words, each search field may impose a property restriction on the search, the property restriction being dependent on the one or more hierarchical object types that are specified.

(19) Example Object-Centric Data Model

(20) FIG. 1 depicts an example object-centric data model. Referring to FIG. 1, object-centric data model **100** includes data objects **102**, **104** and relationships **106A-N**. Data object **102** includes object ID **108**, object type **110**, displayed data **112**, property IDs **114A-N**, and related object IDs **116**. Property ID **114A** is associated with property type **118**, displayed type **120**, and property value **122**. Relationship **106N** is associated with relationship type **124**, displayed type **126**, and related object IDs **128**.

(21) Object-centric data model **100** may be a logical data model that defines how data is represented. Object-centric data model **100** may be independent of any data storage model. For example, data may be stored in a relational database or a key-value store and still be represented by object-centric data model **100**.

(22) At a minimum, object-centric data model **100** is built on the notion of a data object **102**, **104**. A data object **102**, **104** may represent a particular person, a particular location, a particular

organization, a particular event, a particular document, or other instance of a noun. For example, a particular data object may correspond to “Barack Obama”, “San Francisco”, “Stanford University”, “2008 Financial Crisis”, etc.

(23) A data object **102, 104** may be associated with zero or more properties. A property of a data object **102, 104** may be an attribute of the data object **102, 104** and may represent an individual data item. For example, a property may include a name, height, weight, or phone number of a person.

(24) Data objects **102, 104** may be related based on one or more relationships **106A-N**. A relationship **106A-N** may be symmetric or asymmetric. For example, a pair of data objects **102, 104** may be related by an asymmetric “child of” relationship and/or a symmetric “kin of” relationship.

(25) At the highest level of abstraction, a data object **102, 104** may be a container for information. The information may include object ID **108**, object type **110**, displayed data **112**, property IDs **114A-N**, related object IDs **116**, any associated media (e.g., image, video recording, audio recording), any links to associated media, and/or any other data relevant to the data object **102, 104**.

(26) Data objects **102, 104** may be referenced based on unique identifiers that uniquely identify each data object **102, 104**. Thus, a particular data object may store object ID **108**, which is the unique identifier assigned to the particular data object. The particular data object may also store one or more related object IDs **116**, which are the unique identifiers assigned to data objects **102, 104** that share one or more relationships **106A-N** with the particular data object.

(27) Object type **110** may indicate a category of data objects **102, 104**. Example categories include “person”, “location”, “organization”, “event”, and “document”. The category may be indicated by a Uniform Resource Identifier (URI). For example, if a particular data object represents “Barack Obama”, then object type **110** may be “com.palantir.object.person”.

(28) Displayed data **112** may be a user-friendly representation of a particular data object. For example, displayed data **112** may include a user-friendly version of object type **110**, such as “Person”; a commonly used name for the particular data object, such as “Barack Obama”; and/or a thumbnail preview of the particular data object.

(29) If a particular data object is associated with any properties, the particular data object may store property IDs **114A-N**. Each property ID of property IDs **114A-N** may be associated with a property type **118**, a displayed type **120**, and one or more property values **122**.

(30) Property type **118** may indicate a category of properties, and displayed type **120** may be a user-friendly version of property type **118**. For example, if property type **118** is “com.palantir.property.name”, then displayed type **120** may be “Name”. A particular data object may have multiple properties of the same type. For example, a person may have multiple aliases.

(31) Property value **122** may include one or more values of a particular property. Multiple values may correspond to component values. For example, the property value **122** “Barack Obama” may be broken down into component values “Barack” and “Obama”.

(32) If a particular data object stores any related object IDs **116**, a search for matching object identifiers may be performed to determine a relationship **106A-N** with the particular data object. Each relationship of relationships **106A-N** may be associated with a relationship type **124**, a displayed type **126**, and related object IDs **128**.

(33) Relationship type **124** may indicate a category of relationships **106A-N**, and displayed type **126** may be a user-friendly version of relationship type **124**. For example, if relationship type **124** is “com.palantir.relationship.appearsin”, then displayed type **126** may be “Appears In”. In other words, a search of data object **102** may cause displaying of displayed type **126** of relationship **106A** and/or displayed data **112** of data object **104**. Any further data related to data object **104** may be retrieved based on an additional search.

(34) Example Ontology

(35) Object-centric data model **100** may include a hierarchical data type ontology. FIG. 2 depicts an

example ontology. Referring to FIG. 2, ontology **200** includes object types **202A-N**, **212A-N**. Each object type of object types **202A-N**, **212A-N** is associated with property types **204**, **206**, **214**, **216** and one or more relationship types **208**, **210**, **218**, **220**.

(36) Ontology **200** may be a hierarchical organization of object types **202A-N**, **212A-N**; property types **204**, **206**, **214**, **216**; and/or relationship types **208**, **210**, **218**, **220**. In other words, ontology **200** may define which of the property types **204**, **206**, **214**, **216** and/or relationship types **208**, **210**, **218**, **220** correspond to a particular object type. Ontology **200** may be static or dynamic, depending on whether it can be modified by an end user.

(37) In the example of FIG. 2, object types **202A** and **202N** share the same hierarchical level but correspond to different hierarchical object types. For example, object types **202A** and **202N** may correspond to the hierarchical object types “Person” and “Location”, respectively. Different hierarchical object types may be associated with one or more different property types **204**, **206**, **214**, **216** and/or one or more different relationship types **208**, **210**, **218**, **220**. For example, property types **204** may include “Name”, “Height”, “Weight”, and/or any other property types **204** that are relevant to the hierarchical object type “Person”. In contrast, property types **206** may include “Name”, “Latitude”, “Longitude”, and/or any other property types **206** that are relevant to the hierarchical object types “Location”.

(38) In FIG. 2, object types **212A-N** are depicted as descending from object type **202A**. In other words, object types **212A-N** may be sub-types of object type **202A**. For example, object types **212A** and **212N** correspond to the hierarchical object types “Teacher” and “Lawyer”, which are both sub-types of “Person”. Thus, object types **212A-N** may inherit property types **204** and relationship types **208** from object type **202A**. For example, property types **214** may include property types **204** and one or more additional property types **214** that are relevant to the hierarchical object type “Teacher”. However, property types **214** and property types **216** may differ in at least one property type **214**, **216**. For example, property types **214** may include the property type “Grades Taught”, whereas property types **216** may include the property type “Bar Admissions”.

(39) Hierarchical object types that share the same hierarchical level may be disjunctive and may involve disjunctive searches. For example, object types **212A** and **212N** may be disjunctive insofar as a data object **102**, **104** associated with object type **212A** is excluded from association with object type **212N**. Thus, object types **212A** and **212N** may be searched separately.

(40) In contrast, a search of a particular hierarchical object type may be equivalent to searching each sub-type of the particular hierarchical object type. For example, a single search of object type **202A** may be the equivalent of separate searches of object types **212A-N**.

(41) Example User Interface

(42) Searches of highly structured data may be initiated based on input at a user interface. FIG. 3 depicts an example user interface. Referring to FIG. 3, user interface **300** is provided to a user of a client device.

(43) In the example of FIG. 3, user interface **300** is associated with a native mobile application on a smartphone. In FIG. 3, user interface **300** includes a single search field for accepting one or more keywords as input. However, clicking “Advanced” (e.g., via a touch screen) may cause a different user interface to be provided to the user.

(44) Example Plurality of Search Templates

(45) The different user interface may include a plurality of search templates. FIG. 4 depicts a plurality of example search templates. Referring to FIG. 4, search templates **400A-N** are provided to the user.

(46) Search templates **400A-N** translate user intent into search queries. Each search template of search templates **400A-N** specifies a scope of a search of data represented by an object-centric data model **100**. As shall be described in greater detail below, the scope of the search may be defined by a plurality of search fields specified by a selected search template.

(47) Each search template of search templates **400A-N** specifies one or more hierarchical object types within the scope of the search. The one or more hierarchical object types may include disjunctive object types. Additionally or alternatively, the one or more hierarchical object types may include a hierarchical object type as well as a sub-type of the hierarchical object type.

(48) For example, in FIG. 4, search template **400A** corresponds to the hierarchical object type “Person”. As mentioned above, a search of a particular object type may be equivalent to searching each sub-type of the particular object type. Thus, any sub-type of a hierarchical object type that is within the scope of the search is also within the scope of the search. For example, the hierarchical object types “Teacher” and “Lawyer” are also within the scope of the search specified by search template **400A**. In other words, a “Person” search may be implemented as separate searches of “Teacher” and “Lawyer” that are performed disjunctively.

(49) In another example, a user may modify the plurality of search templates **400A-N** of FIG. 4 to include a customized search template that specifies the hierarchical object types “Person” and “Location”. The customized search template may be generated by editing an existing search template or creating a new search template. Thus, the customized search template may specify disjunctive searches of “Person” and “Location”.

(50) Approaches for Obtaining Input to Search Fields

(51) Selecting a particular search template may cause generating a user interface **300** that includes a plurality of search fields. For example, FIG. 5A depicts an example plurality of search fields **500A-N** provided to the user in response to selecting a particular search template. In an embodiment, a search field may accept a keyword and/or a property value **122** as input.

(52) Each search template of search templates **400A-N** may specify search fields **500A-N** that correspond to relevant property types associated with one or more data objects **102**, **104** that are within the scope of the search defined by the search template. In the example of FIG. 5A, the particular search template may be a customized search template specifying disjunctive searches for a “Person” and a “Document”. Thus, search fields **500A-N** may correspond to relevant property types associated with the hierarchical object types “Person” and “Document”.

(53) Each search field of the plurality of search fields **500A-N** may be associated with a displayed type **120**. For example, search field **500A** follows the displayed type **120** “First”. Furthermore, each search field of the plurality of search fields **500A-N** accepts input specifying a property value **122**. For example, search field **500A** accepts input specifying the property value **122** “John”. Note that input need not be provided to all of the search fields **500A-N**. Furthermore, there may be hidden search fields associated with fixed property values, such as “USA” in a hidden “Country” search field.

(54) Referring to FIG. 5A, there is a “Properties” section that includes search field **500A** (hereinafter “property filter”). A property filter may specify a conjunctive search for each property type **118** that is provided with a property value **122**. For example, a conjunctive search may be performed for the first name “John” and the last name “Smith”.

(55) FIG. 5A also depicts a “Date Range” section that includes search field **500N** (hereinafter “intrinsic date search”). An intrinsic date search is typically used with the hierarchical object types “Document” and “Event”. An “intrinsic date” may be a date of creation, a date of occurrence, a date of publication, or any other property that can be represented as a date.

(56) A particular search field may obtain input in any of a number of different ways. Each search field of the plurality of search fields **500A-N** is associated with an input type (e.g., SIMPLE, DATE, NUMERIC, ENUM, MAP). Thus, search fields **500A-N** may include a text entry field, a date picker, a numeric keypad, a selectable list, an interactive map, or any other interface for obtaining input. FIG. 5B depicts a user interface **300** that obtains input in multiple ways.

(57) Referring to FIG. 5B, each search field of search fields **502** corresponds to the same property type “First”. Selecting a “+” button may add an additional search field to search fields **502**. Search fields **502** are disjunctive search fields. In other words, each property value **122** provided to search

fields **502** may be searched disjunctively. However, search fields **502** may still be searched conjunctively with any other search fields.

(58) FIG. **5B** also depicts input **504** obtained for the property type “Location”. Input **504** may have been obtained based on selecting the crosshairs button adjacent to the “Location” search field. The crosshairs button may have caused the user to be provided with the interface depicted in FIG. **5C**.

(59) FIG. **5C** depicts an example map interface including an interactive map **506**. The user may add a geolocation pin to the interactive map **506**. The geolocation pin may indicate a particular location and/or the center of a geographical region relevant to a search. For example, a geo-fenced search may be performed based on a user-specified geographical region **508**. Thus, a particular search template may restrict a search to one or more locations. For example, the particular search template may restrict a scope of a search to one or more geographic regions.

(60) In an embodiment (not shown), an input **504** to a particular search field may be obtained from a camera and/or any other hardware of the client computer at which the user interface **300** is displayed. For example, a camera may be used to populate search fields with measurements (e.g., distance between eyes, length of face, width of face) that can be used for face recognition.

(61) Example Customizable Format for Search Results

(62) After a search is performed, search results may be retrieved and provided to the user. The search results may be provided in any number of different formats, which may be customized by the user. FIG. **6** depicts an example customizable format for search results.

(63) Referring to FIG. **6**, customizable format **600** includes a list of search results. The list may be arranged in any order. For example, the list may be organized based on hierarchical object type.

(64) Search results may be presented in any of a number of different ways. Each search result may include some or all of the displayed data **112** associated with the search result. Additionally or alternatively, one or more search results may be plotted on a map. Additionally or alternatively, the search results may be represented graphically using any of the techniques described in U.S. patent application Ser. No. 13/608,864, filed Sep. 10, 2012, the entirety of which is incorporated herein by reference. Additionally or alternatively, the search results may be provided as selectable histograms using any of the techniques described in U.S. patent application Ser. No. 14/676,621, filed Apr. 1, 2015, the entirety of which is incorporated herein by reference.

(65) In addition to the search results, the user may be provided with past search results. For example, the results of recent searches may be provided to the user as a list. Current and/or past search results may be used to modify search templates **400A-N**.

(66) Example Detailed Views of a Search Result

(67) Selecting a particular search result may cause the user to be provided with one or more detailed views of the particular search result. FIGS. **7-9** depict example detailed views of a search result.

(68) FIG. **7** depicts an example “Profile” view. This view may include a summary of properties associated with a particular data object. In the example of FIG. **7**, this view includes displayed data **112** associated with the particular data object as well as displayed type **120** and property value **122** for each property of the particular data object.

(69) The user can do any of a number of things with a particular data object obtained as a search result. The user can edit the particular data object. For example, the user can modify a property and store the modified data object in the data store from which the unmodified data object was retrieved. Additionally or alternatively, the user can share the particular data object with another user. For example, the user may send a link to the particular data object to another user.

Additionally or alternatively, the user can specify additional searches to be performed based on the particular data object using any of the techniques described in U.S. patent application Ser. No. 13/608,864. Additionally or alternatively, a search result may be geotagged (e.g., associated with a particular location).

(70) FIG. **8** depicts an example “Related” view. This view may include any data objects **102**, **104**

that are related to the particular data object obtained as a search result. In the example of FIG. 8, this view includes displayed type **126** of each relationship between the particular data object and a related data object as well as the displayed data **112** associated with the related data object.

(71) FIG. 9 depicts an example “Media” view. This view may include links to any media data associated with the particular data object obtained as a search result. Selecting a media link may cause media data to be rendered in a suitable manner. For example, selecting a video file may cause the video file to be presented in a video player.

(72) Process Overview

(73) FIG. 10 is a flow diagram that depicts an approach for searching highly structured data. At block **1000**, an indication of a search template is received. The indication may include user input (e.g., input indicating a user's selection of a search template in a user interface) and/or a network message (e.g., a HTTP message that indicates a user's selection of a search template). The indication may specify that the search template is to be used for a search of data represented by an object-centric data model. The search template may specify the scope of the search, which may include one or more hierarchical object types that are defined in the object-centric data model. The search template may also specify a plurality of search fields.

(74) For example, a user may select a “Person” search template. The “Person” search template may specify that a search is to be performed for data belonging to the “Person” object type. A “Person” object type is associated with certain property types. Thus, selecting the “Person” search template specifies these property types.

(75) At block **1002**, a user interface is generated based on the search template. The user interface may include the plurality of search fields. For example, the property types specified by the “Person” search template may translate into search fields corresponding to a person's name, age, address, phone number, etc.

(76) At optional block **1004**, the user interface is provided to a user. For example, search fields may be provided to the user so that the user can input property values to be matched when the search is performed.

(77) At optional block **1006**, input is received from the user through the user interface. The input may be in any of a number of formats. For example, the input may be a text entry, an interaction with a map, a selection from an enumerated list, etc.

(78) At optional block **1008**, the search is performed based on the input. For example, text input may be enclosed with wildcard operators, such as “*”, and matched against stored property values.

(79) At optional block **1010**, one or more first data objects are retrieved as a result of performing the search. The one or more first data objects are data objects that fell within the scope of the search. Any of a number of subsequent actions may be performed based on the one or more first data objects, including searching for one or more second data object that did not fall within the scope of the search.

(80) Example Computer System

(81) FIG. 11 depicts an example computer system in which embodiments may be implemented. Referring to FIG. 11, client computer **1100** includes client application **1102**. Client computer **1100** is communicatively coupled to server computer **1104**, which is communicatively coupled to body of data **1106**.

(82) A “computer” may be one or more physical computers, virtual computers, and/or computing devices. As an example, a computer may be one or more server computers, cloud-based computers, cloud-based cluster of computers, virtual machine instances or virtual machine computing elements such as virtual processors, storage and memory, data centers, storage devices, desktop computers, laptop computers, mobile devices, and/or any other special-purpose computing devices. A computer may be a client and/or a server. Any reference to “a computer” herein may mean one or more computers, unless expressly stated otherwise.

(83) Client application **1102** may be a sequence of instructions executing at client computer **1100**.

At a minimum, client application **1102** may provide, to a user, any of the interfaces described in FIGS. 2-9. For example, client application **1102** may be a program executing in a web browser or a native mobile app. Client application **1102** may be installed on client computer **1100** to provide any of a number of benefits. The benefits may include faster execution, lower network latency and bandwidth consumption, and/or better access to the hardware of client computer **1100**. Furthermore, a client application **1102** installed on client computer **1100** enables the client application **1102** to operate in an offline mode. For example, search results may be saved locally on client computer **1100** so that a user can interact with them without an Internet connection.

(84) Server computer **1104** may include one or more computers, such as a web server, a mobile server, a gateway server, and/or a load-balancing server. A gateway server may regulate access to other servers, including structured data servers and unstructured data servers. Structured data servers may be computers that facilitate searches of structured data, such as data stored in an object-centric data model. Unstructured data servers may be computers that facilitate searches of unstructured data, such as by implementing text searches of documents. Server computer **1104** may send data to client computer **1100** using JavaScript Object Notation (JSON), Extensible Markup Language (XML), and/or any other data interchange format.

(85) Body of data **1106** may represent the data being searched. For example, body of data **1106** may be stored in a database, a configuration file, and/or any other system and/or data structure that stores data. Additionally or alternatively, body of data **1106** may be stored in memory on server computer **1104**. Additionally or alternatively, body of data **1106** may be stored in non-volatile storage. For example, body of data **1106** may be stored in a mobile database communicatively coupled to a mobile server and/or a repository communicatively coupled to a gateway server.

(86) A user's access to body of data **1106** may be limited based on access controls. Thus, a search of a body of data **1106** may be restricted to data to which the user has access.

(87) Basic Computing Device

(88) Referring now to FIG. 12, it is a block diagram that illustrates a basic computing device **1200** in which the example embodiment(s) of the present Application may be embodied. Computing device **1200** and its components, including their connections, relationships, and functions, is meant to be exemplary only, and not meant to limit implementations of the example embodiment(s). Other computing devices suitable for implementing the example embodiment(s) may have different components, including components with different connections, relationships, and functions.

(89) Computing device **1200** may include a bus **1202** or other communication mechanism for addressing main memory **1206** and for transferring data between and among the various components of device **1200**.

(90) Computing device **1200** may also include one or more hardware processors **1204** coupled with bus **1202** for processing information. A hardware processor **1204** may be a general purpose microprocessor, a system on a chip (SoC), or other processor.

(91) Main memory **1206**, such as a random access memory (RAM) or other dynamic storage device, also may be coupled to bus **1202** for storing information and software instructions to be executed by processor(s) **1204**. Main memory **1206** also may be used for storing temporary variables or other intermediate information during execution of software instructions to be executed by processor(s) **1204**.

(92) Software instructions, when stored in storage media accessible to processor(s) **1204**, render computing device **1200** into a special-purpose computing device that is customized to perform the operations specified in the software instructions. The terms “software”, “software instructions”, “computer program”, “computer-executable instructions”, and “processor-executable instructions” are to be broadly construed to cover any machine-readable information, whether or not human-readable, for instructing a computing device to perform specific operations, and including, but not limited to, application software, desktop applications, scripts, binaries, operating systems, device drivers, boot loaders, shells, utilities, system software, JAVASCRIPT, web pages, web applications,

plugins, embedded software, microcode, compilers, debuggers, interpreters, virtual machines, linkers, and text editors.

(93) Computing device **1200** also may include read only memory (ROM) **1208** or other static storage device coupled to bus **1202** for storing static information and software instructions for processor(s) **1204**.

(94) One or more mass storage devices **1210** may be coupled to bus **1202** for persistently storing information and software instructions on fixed or removable media, such as magnetic, optical, solid-state, magnetic-optical, flash memory, or any other available mass storage technology. The mass storage may be shared on a network, or it may be dedicated mass storage. Typically, at least one of the mass storage devices **1210** (e.g., the main hard disk for the device) stores a body of program and data for directing operation of the computing device, including an operating system, user application programs, driver and other support files, as well as other data files of all sorts.

(95) Computing device **1200** may be coupled via bus **1202** to display **1212**, such as a liquid crystal display (LCD) or other electronic visual display, for displaying information to a computer user. In some configurations, a touch sensitive surface incorporating touch detection technology (e.g., resistive, capacitive, etc.) may be overlaid on display **1212** to form a touch sensitive display for communicating touch gesture (e.g., finger or stylus) input to processor(s) **1204**.

(96) An input device **1214**, including alphanumeric and other keys, may be coupled to bus **1202** for communicating information and command selections to processor **1204**. In addition to or instead of alphanumeric and other keys, input device **1214** may include one or more physical buttons or switches such as, for example, a power (on/off) button, a “home” button, volume control buttons, or the like.

(97) Another type of user input device may be a cursor control **1216**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **1204** and for controlling cursor movement on display **1212**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane.

(98) While in some configurations, such as the configuration depicted in FIG. **12**, one or more of display **1212**, input device **1214**, and cursor control **1216** are external components (i.e., peripheral devices) of computing device **1200**, some or all of display **1212**, input device **1214**, and cursor control **1216** are integrated as part of the form factor of computing device **1200** in other configurations.

(99) Functions of the disclosed systems, methods, and modules may be performed by computing device **1200** in response to processor(s) **1204** executing one or more programs of software instructions contained in main memory **1206**. Such software instructions may be read into main memory **1206** from another storage medium, such as storage device(s) **1210**. Execution of the software instructions contained in main memory **1206** cause processor(s) **1204** to perform the functions of the example embodiment(s).

(100) While functions and operations of the example embodiment(s) may be implemented entirely with software instructions, hard-wired or programmable circuitry of computing device **1200** (e.g., an ASIC, a FPGA, or the like) may be used in other embodiments in place of or in combination with software instructions to perform the functions, according to the requirements of the particular implementation at hand.

(101) The term “storage media” as used herein refers to any non-transitory media that store data and/or software instructions that cause a computing device to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, non-volatile random access memory (NVRAM), flash memory, optical disks, magnetic disks, or solid-state drives, such as storage device **1210**. Volatile media includes dynamic memory, such as main memory **1206**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid-state drive, magnetic tape, or any other

magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, flash memory, any other memory chip or cartridge.

(102) Storage media is distinct from but may be used in conjunction with transmission media.

Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **1202**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(103) Various forms of media may be involved in carrying one or more sequences of one or more software instructions to processor(s) **1204** for execution. For example, the software instructions may initially be carried on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the software instructions into its dynamic memory and send the software instructions over a telephone line using a modem. A modem local to computing device **1200** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **1202**. Bus **1202** carries the data to main memory **1206**, from which processor(s) **1204** retrieves and executes the software instructions. The software instructions received by main memory **1206** may optionally be stored on storage device(s) **1210** either before or after execution by processor(s) **1204**.

(104) Computing device **1200** also may include one or more communication interface(s) **1218** coupled to bus **1202**. A communication interface **1218** provides a two-way data communication coupling to a wired or wireless network link **1220** that is connected to a local network **1222** (e.g., Ethernet network, Wireless Local Area Network, cellular phone network, Bluetooth wireless network, or the like). Communication interface **1218** sends and receives electrical, electromagnetic, or optical signals that carry digital data streams representing various types of information. For example, communication interface **1218** may be a wired network interface card, a wireless network interface card with an integrated radio antenna, or a modem (e.g., ISDN, DSL, or cable modem).

(105) Network link(s) **1220** typically provide data communication through one or more networks to other data devices. For example, a network link **1220** may provide a connection through a local network **1222** to a host computer **1224** or to data equipment operated by an Internet Service Provider (ISP) **1226**. ISP **1226** in turn provides data communication services through the world wide packet data communication network now commonly referred to as the “Internet” **1228**. Local network(s) **1222** and Internet **1228** use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link(s) **1220** and through communication interface(s) **1218**, which carry the digital data to and from computing device **1200**, are example forms of transmission media.

(106) Computing device **1200** can send messages and receive data, including program code, through the network(s), network link(s) **1220** and communication interface(s) **1218**. In the Internet example, a server **1230** might transmit a requested code for an application program through Internet **1228**, ISP **1226**, local network(s) **1222** and communication interface(s) **1218**.

(107) The received code may be executed by processor **1204** as it is received, and/or stored in storage device **1210**, or other non-volatile storage for later execution.

(108) Basic Software System

(109) FIG. **13** is a block diagram of a basic software system **1300** that may be employed for controlling the operation of computing device **1200**. Software system **1300** and its components, including their connections, relationships, and functions, is meant to be exemplary only, and not meant to limit implementations of the example embodiment(s). Other software systems suitable for implementing the example embodiment(s) may have different components, including components with different connections, relationships, and functions.

(110) Software system **1300** is provided for directing the operation of computing device **1200**.

Software system **1300**, which may be stored in system memory (RAM) **1206** and on fixed storage (e.g., hard disk or flash memory) **1210**, includes a kernel or operating system (OS) **1310**.

(111) The OS **1310** manages low-level aspects of computer operation, including managing execution of processes, memory allocation, file input and output (I/O), and device I/O. One or more application programs, represented as **1302A**, **1302B**, **1302C** . . . **1302N**, may be “loaded” (e.g., transferred from fixed storage **1210** into memory **1206**) for execution by the system **1300**. The applications or other software intended for use on device **1300** may also be stored as a set of downloadable computer-executable instructions, for example, for downloading and installation from an Internet location (e.g., a Web server, an app store, or other online service).

(112) Software system **1300** includes a graphical user interface (GUI) **1315**, for receiving user commands and data in a graphical (e.g., “point-and-click” or “touch gesture”) fashion. These inputs, in turn, may be acted upon by the system **1300** in accordance with instructions from operating system **1310** and/or application(s) **1302**. The GUI **1315** also serves to display the results of operation from the OS **1310** and application(s) **1302**, whereupon the user may supply additional inputs or terminate the session (e.g., log off).

(113) OS **1310** can execute directly on the bare hardware **1320** (e.g., processor(s) **1204**) of device **1200**. Alternatively, a hypervisor or virtual machine monitor (VMM) **1330** may be interposed between the bare hardware **1320** and the OS **1310**. In this configuration, VMM **1330** acts as a software “cushion” or virtualization layer between the OS **1310** and the bare hardware **1320** of the device **1200**.

(114) VMM **1330** instantiates and runs one or more virtual machine instances (“guest machines”). Each guest machine comprises a “guest” operating system, such as OS **1310**, and one or more applications, such as application(s) **1302**, designed to execute on the guest operating system. The VMM **1330** presents the guest operating systems with a virtual operating platform and manages the execution of the guest operating systems.

(115) In some instances, the VMM **1330** may allow a guest operating system to run as if it is running on the bare hardware **1320** of device **1200** directly. In these instances, the same version of the guest operating system configured to execute on the bare hardware **1320** directly may also execute on VMM **1330** without modification or reconfiguration. In other words, VMM **1330** may provide full hardware and CPU virtualization to a guest operating system in some instances.

(116) In other instances, a guest operating system may be specially designed or configured to execute on VMM **1330** for efficiency. In these instances, the guest operating system is “aware” that it executes on a virtual machine monitor. In other words, VMM **1330** may provide para-virtualization to a guest operating system in some instances.

(117) The above-described basic computer hardware and software is presented for purpose of illustrating the basic underlying computer components that may be employed for implementing the example embodiment(s). The example embodiment(s), however, are not necessarily limited to any particular computing environment or computing device configuration. Instead, the example embodiment(s) may be implemented in any type of system architecture or processing environment that one skilled in the art, in light of this disclosure, would understand as capable of supporting the features and functions of the example embodiment(s) presented herein.

(118) Extensions and Alternatives

(119) In the foregoing specification, the example embodiment(s) of the present Application have been described with reference to numerous specific details. However, the details may vary from implementation to implementation according to the requirements of the particular implement at hand. The example embodiment(s) are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A computer-implemented method, comprising: receiving a selection of a particular search template from a plurality of search templates to be used for a computer-implemented search of digital data stored within an object-centric data model, the object-centric data model defining data objects and relationships between data objects and being based on a hierarchical ontology having hierarchical object types with associated property types, the particular search template specifying one or more hierarchical object types of the object-centric data model that define a scope of the search and at least one search field related to at least one object type of the one or more hierarchical object types, each search field of the particular search template imposing a property restriction on the search, the property restriction being dependent on the one or more hierarchical object types that are specified, the at least one search field including one or more search fields that accept input from an interactive map, two different search templates of the plurality of search templates having respective search fields corresponding to different sets of object types that include a common object type; based on the particular search template, generating a user interface for an electronic visual display, the user interface representing the at least one search field for the search as at least one input area, the user interface allowing adding a specific input area based on the at least one search field related to the at least one object type; in response to receiving a request including an object of the at least one object type to perform a search through the user interface that includes the specific input area, performing a search within the scope based on the request including searching each sub-type of the at least one object type; retrieving a result of the search including one or more first data objects from the object-centric data model; and providing, in the user interface, the one or more first data objects that are geotagged.
2. The method of claim 1, wherein the at least one input area includes one or more input areas that accept input from the interactive map.
3. The method of claim 1, wherein the particular search template restricts the search to a user-specified geographical region.
4. The method of claim 1, wherein a particular property type associated with the one or more hierarchical object types corresponds to two or more disjunctive search fields in the at least one search field.
5. The method of claim 1, wherein the one or more hierarchical object types include two or more disjunctive object types.
6. The method of claim 1, further comprising providing one or more second data objects that are outside the scope of the search based on one or more relationships, defined in the object-centric data model, between the one or more first data objects and the one or more second data objects.
7. The method of claim 1, wherein the particular search template specifies the scope of the search.
8. The method of claim 1, further comprising providing the user interface to the electronic visual display subsequent to generating the user interface.
9. The method of claim 1, wherein the one or more hierarchical object types include two or more disjunctive object types.
10. A computer system, comprising: one or more processors; and one or more non-transitory data storage media storing one or more programs that are configured for execution by the one or more processors, the one or more programs comprising one or more sequences of instructions which when executed using the one or more processors cause the one or more processors to perform: receiving a selection of a particular search template from a plurality of search templates to be used for a computer-implemented search of digital data stored within an object-centric data model, the object-centric data model defining data objects and relationships between data objects and being based on a hierarchical ontology having hierarchical object types with associated property types, the particular search template specifying one or more hierarchical object types of the object-centric data model that define a scope of the search and at least one search field related to at least one object type of the one or more hierarchical object types, each search field of the particular search

template imposing a property restriction on the search, the property restriction being dependent on the one or more hierarchical object types that are specified, the at least one search field including one or more search fields that accept input from an interactive map, two different search templates of the plurality of search templates having respective search fields corresponding to different sets of object types that include a common object type; based on the particular search template, generating a user interface for an electronic visual display, the user interface representing the at least one search field for the search as at least one input area, the user interface allowing adding a specific input area based on the at least one search field related to the at least one object type; in response to receiving a request including an object of the at least one object type to perform a search through the user interface that includes the specific input area, performing a search within the scope based on the request including searching each sub-type of the at least one object type; retrieving a result of the search including one or more first data objects from the object-centric data model; and providing, in the user interface, the one or more first data objects that are geotagged.

11. The computer system of claim 10, wherein the at least one input area includes one or more input areas that accept input from the interactive map.

12. The computer system of claim 10, wherein the particular search template restricts the search to a user-specified geographical region.

13. The computer system of claim 10, wherein a particular property type associated with the one or more hierarchical object types corresponds to two or more disjunctive search fields in the at least one search field.

14. The computer system of claim 10, the one or more sequences of instructions which when executed using the one or more processors causing the one or more processors to perform providing one or more second data objects that are outside the scope of the search based on one or more relationships, defined in the object-centric data model, between the one or more first data objects and the one or more second data objects.

15. The computer system of claim 10, wherein the particular search template specifies the scope of the search.

16. The computer system of claim 10, further comprising sequences of instructions which when executed cause providing the user interface to the electronic visual display subsequent to generating the user interface.
